

SMART INDIA HACKATHON 2026



SMART INDIA
HACKATHON
2026

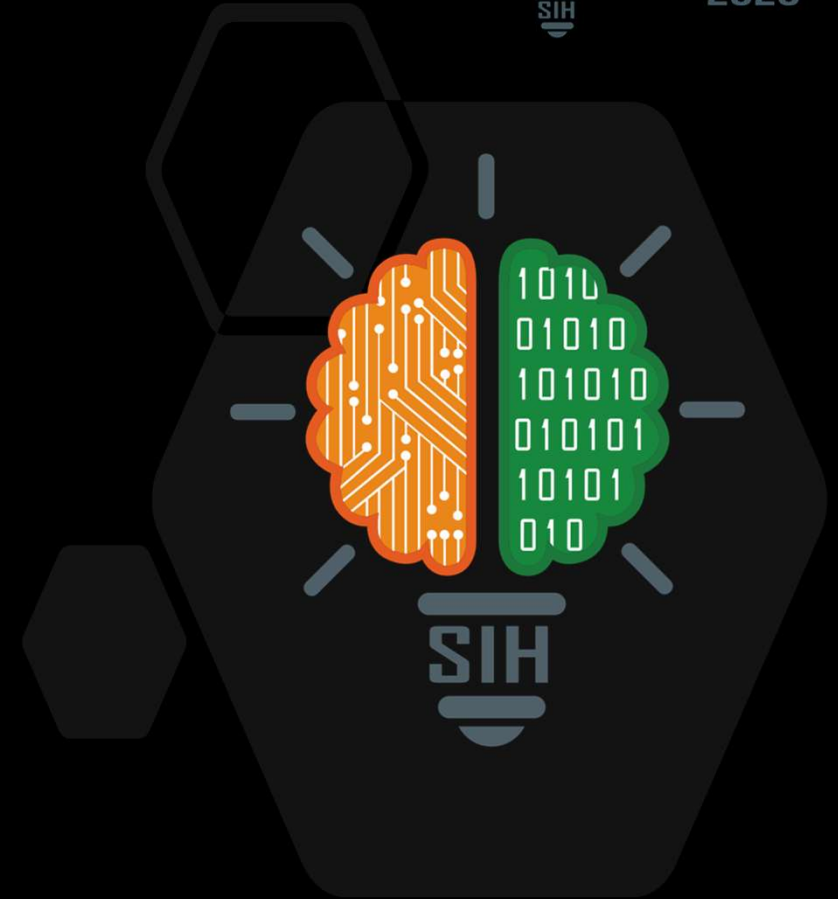
Problem Statement ID (PS id) - SIH26194

Problem Statement Title - Student Innovation

Theme - Fitness & Sports

PS Category - Software

Team Name - AlgoCrew



❖ Proposed Solution

- Record / Upload: Boxers record or upload training videos through the Android app.
- AI Analysis: Computer vision detects punches, stance, movement, guard and technique.
- Performance Feedback: The app provides technique scores, identifies weaknesses and gives improvement suggestions.
- Personalized Training: Analysis is used to track progress and recommend areas to focus on.

Key Innovation:

Affordable, mobile-based AI coaching without specialized equipment.



Technologies

- **Android:** Kotlin + Jetpack Compose
- **AI/CV:** Python + OpenCV + Pose Estimation
- **Backend:** FastAPI
- **Database:** Firebase
- **AI Model:** Punch & movement classification

System Flow

Record / Upload Video -> Pose & Movement Detection -> Punch / Technique Recognition -> Performance Analysis -> Feedback & Improvement Suggestions -> Results shown in Android App



Technical

- Uses smartphone video + existing AI/CV models
- Cloud-based processing

Challenges

- Fast movement detection
- Varying camera angles & lighting
- Quality training data

Scalability

- Low-cost and hardware-independent
- Can expand beyond boxing

Strategies for Overcoming Challenges

- Fast movements: Pose estimation + frame-by-frame video analysis
- Camera variations: Training with diverse angles, lighting and backgrounds
- Limited data: Use publicly available datasets + augmented training data

For Athletes

- Personalized feedback & progress tracking
- Identification of technical weaknesses

For Coaches

- Faster, objective performance analysis
- Data-driven training decisions

Broader Benefits

- Social: Makes advanced sports analysis more accessible
- Economic: Reduces dependence on expensive analysis equipment
- Environmental: Reduces need for specialized physical equipment

Scalability

- Adaptable to other sports



MediaPipe Pose — Real-time human pose estimation

YOLO — Real-time object detection

OpenCV — Computer vision and video processing

Sports Video Analytics Research — Computer vision for athlete movement and performance analysis

Action Recognition Research — Temporal modeling of movement sequences

Key technologies researched:

Computer Vision • Pose Estimation • Action Recognition • Temporal ML